

6. Auxiliary Equipment**6.1 Y Table for Roughing Mill to feed Top passes automatic**

Purpose : To receive the 1st & 3rd Pass and convey the material to Y troughs to feed to top passes of mill stands

Quantity : One (1) Set

Length : 10000 mm Appx.

Roller Spacing : 1000mm

Rollers

- Diameter : 268mm

- Barrel Length : 1200 mm

No. of Rollers : Nine (9) Nos.

Speed : 3.0 m/sec

Shaft Bearing : Spherical Roller Bearings

Frame : Fabricated from steel structure

Drive : Group driven with chain and sprockets through worm reducer

Motor kw : 11 KW

Motor RPM : 1000 RPM

Material of construction

Frame : Mild Steel Construction of E 250 as per IS : 2062

Sprocket : 45C8 as per IS:1570 Induction hardened to 250-280 BHN

Shaft : 40C8 as per IS : 1570

Rollers : Seamless Pipes



6.2 Fix Roller Table for Roughing Mill to feed Bottom passes automatic

Purpose	:	To Feed Bottom Passed on Roughing Mill
Quantity	:	One (1) Set
Length	:	8000 mm Appx.
Roller Spacing	:	1000mm
Rollers		
- Diameter	:	268mm
- Barrel Length	:	1200 mm
No. of Rollers	:	Nine (9) Nos.
Speed	:	3.0 m/sec
Shaft Bearing	:	Spherical Roller Bearings
Frame	:	Fabricated from steel structure
Drive	:	Group driven with chain and sprockets through worm reducer
Motor kw	:	11 KW
Motor RPM	:	1000 RPM

Material of construction

Frame	:	Mild Steel Construction of E 250 as per IS : 2062
Sprocket	:	45C8 as per IS:1570 Induction hardened to 250-280 BHN
Shaft	:	40C8 as per IS : 1570
Rollers	:	Seamless Pipes
Apron Plate	:	Minimum 20 mm Fabricated from Mild Steel Plates



6.3 Approach Roller Table to 1st Continuous Mill (Extension Roller Table)

Purpose	:	To handle the material after roller table with flappers
Total Length	:	15000 mm Appx.
Table Length (C/C to end rollers)	:	15,000 mm
Roller Spacing	:	1000 mm
Rollers		
- Diameter	:	246 mm
- Barrel Length	:	700 mm
No. of Rollers	:	Sixteen (16) Nos.
Shaft Bearing	:	Spherical Roller Bearings - 22220
Roller Drive	:	Group driven with Chain & Sprocket through suspended gearbox
Motor KW	:	11 KW
Overall reduction ratio	:	4.5
Motor RPM	:	1000 RPM
Peripheral Speed of Rollers	:	3.0 m/s
Material of construction		
Frame	:	Mild Steel Construction of E 250 as per IS : 2062
Sprocket	:	45C8 as per IS:1570 Induction hardened to 250-280 BHN
Shaft	:	40C8 as per IS: 1570
Rollers	:	Seamless Pipes
Apron Plate	:	Minimum 20mm Fabricated from Mild Steel Plates



6.4 Pinch roll Before Quenching Box

Purpose	:	To pinch and convey the high-speed bars
Location	:	In between TMT Line
Quantity	:	One (1) No.
Type	:	Cantilever
Pinch Roll Dia	:	Max. Ø 310 mm Min. Ø 270 mm
Barrel Length	:	Pinch Roll - 70mm
Bar Speed	:	12 m/sec-22 m/sec
Motor	:	89 KW
RPM	:	1500 RPM
Drive	:	With reduction gearbox and cardan shaft (top roll pneumatically operated)
Ratio	:	As per requirement
Pinching	:	Pneumatic by operation of Top Roll
Lubrication	:	Splash Oil/Forced
Bearings	:	SKF/FAG Make
Body Construction	:	Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	:	Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	:	Flexible Gear Coupling
Gears Material	:	18CrNiMo7-6 case hardened & heat treated
Shafts Material	:	18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	:	Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings



6.5 Pinch roll after Quenching Box

Purpose	:	To pinch and convey the high-speed bars
Location	:	In between TMT Line
Quantity	:	One (1) No.
Type	:	Cantilever
Pinch Roll Dia	:	Max. Ø 310 mm
Min. Ø 270 mm		
Barrel Length	:	Pinch Roll - 70mm
Bar Speed	:	12 m/sec-22 m/sec
Motor	:	89 KW
RPM	:	1500 RPM
Drive	:	With reduction gearbox and cardan shaft (top roll pneumatically operated)
Ratio	:	As per requirement
Pinching	:	Pneumatic by operation of Top Roll
Lubrication	:	Splash Oil/Forced
Bearings	:	SKF/FAG Make
Body Construction	:	Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	:	Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	:	Flexible Gear Coupling
Gears Material	:	18CrNiMo7-6 case hardened & heat treated
Shafts Material	:	18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	:	Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings



6.6 Start/ Stop type Crop & Cobble Shear – 2 Nos.

For Hot Material Cutting (Temp. 850°C to 1050°C)

Type : Direct Driven Dividing Shear (Steel Fabricated)

Motor : DC Motor with drive & Base Plate

Low Speed Start / Stop Shear with Lubrication Piping, Oil Supply System,
Delivery Guide Trough

Gears : Case Hardened & Ground with Surface Hardness HRC 58-60

Shafts : Heat Treated Steel

Blades : Heat Treated Steel (46-50 HRC)

Maximum Cross Sectional Area of the bar to : 3200 mm sq to be cut

Design Shear Strength of Bar at operating : 125 N/mm² temperature

Design Cutting Force : 400KN

Bar Speed : 0.8 - 1.3m/s

Each Shear complete with:

- Lubricating Piping on machine suitable for connection to central oil supply system.
- Entry & Exit Guides
- Crop Collection System including Steel fabricated chutes & buckets, pneumatic drop Guide



6.7 Tail Breaking Pinch Rolls**– 2 Nos.**

Purpose	: To pinch and feed the material to Twin channels
Location	: At entry of twin channels
Type	: Cantilever
Pinch Roll Diameter	: Max. Ø 300mm Min. Ø270mm
Barrel Length	: 70 mm
Bar Speed	: 12m/sec -22m/sec
Motor Power	: 89 KW
Motor Speed	: 1500 RPM
Drive	: With reduction gearbox and cardon shaft (top roll pneumatically operated)
Ratio	: As per requirement
Pinching	: Pneumatic by operation of Top Roll
Shifting of Pinch Rolls	: Hydraulic by operation
Lubrication	: Forced oil lubrication from Central Oil Lubrication System # 3.
Bearings	: Structural, SKF/FAG Make
Body Construction	: Structural Steel Fabricated, duly stress relieved & machined
Common Base Frame for Pinch Roll & Motor	: Structural Steel Fabricated, duly stress relieved & machined
Drive Coupling	: Flexible Gear Coupling
Gears Material	: 18CrNiMo7-6 case hardened & heat treated
Shafts Material	: 18CrNiMo7-6 case hardened & heat treated
Entry & Exit Guides	: Cast Steel/Steel Fabricated

Complete with pneumatic equipment comprising FRL unit, directional valves, pipes, hoses & fittings

6.8 Roller Guide Box with Holder**1 Set****6.9 Roller Twist Guide****1 Set****6.10 Funnels****1 Set****6.11 Static Guides****1 Set****6.12 Water pump for Rolling Mill
(1 Working + 1 Stand by)****2 Nos.**